



Downstairs resident classification characteristics for upstairs walking vibration noise in an apartment building under virtual reality environment

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ABSTRACT

Psychoacoustic experiments were performed to investigate head-related transfer function (HRTF) and head-mounted display (HMD) effects, to develop a new virtual-reality (VR)-based noise assessment methodology and classification method for heavy-weight impact sounds in an apartment indoor environment. Sound pressure levels and frequency characteristics before and after HRTF application were examined to reproduce the “upstairs condition” (overhead movement), for a heavy-weight impact source recorded in an apartment living room. Floor impact noise was assessed using the HMD by modeling a typical living room. For four environments (NONE, HRTF, HMD, and HRTF + HMD), subjective responses (annoyance, allowance limit, semantic expression) were examined for 38–62-dB maximum sound pressure levels with an A-weighted frequency response and fast time constant. Directional information through the HRTF significantly increased annoyance and decreased the allowance limit, and HMD visual information partially (1.2 times) affected the allowance limit in a high inter-floor noise environment owing to increased visual involvement (56 dBA). The semantic expression assessment revealed comprehensive inter-floor noise responses of *dissatisfaction*, *irregularity*, and *discontinuity*, and confirmed greater annoyance due to walking vibration noise in an environment with directional information, because of increased perception of heavy-impact sound-induced discomfort. Regarding annoyance- and allowance-limit-based classification, the satisfaction standards for low-noise environments were assessed as 6–7 dB lower than those of previous studies when both HRTF and HMD were applied, confirming that test subjects respond to noise more sensitively in a virtual environment where audiovisual information is provided. Therefore, a new indoor noise assessment methodology for environmental factors is proposed herein.

1. Introduction

In the sound environments of apartment buildings, residents can easily be exposed to noise from upstairs and neighboring households, which is transferred through ceilings, floors, and walls. In South Korea, more than 70% of apartment residents suffer floor impact sound noise, such as that from running and jumping children and moving furniture [1,2]. In particular, 73% of complaints related to floor impact sounds are caused by heavy-weight impact noise, such as that from adult footsteps and running and jumping children [3]. Standards for floor impact sound measurement have been developed, including International Organization for Standardization (ISO) 10140-3 (using a rubber ball and tapping machine) [4], Korean Industrial Standard (KS) 2810 [5], and Japanese Industrial Standard (JIS) 1418 (using a rubber ball, bang machine, and tapping machine) [6]. However, reproduction of the walking, running, or jumping vibration noise that causes serious disputes is difficult. Therefore, actual and standard noise sources have

been compared in various studies; hence, similarity between actual noise sources and a dropping rubber ball in terms of impact sound level and tone has been determined [1,7].

As with general noise, the annoyance caused by inter-floor noise is significantly affected by loudness, and the loudness of the perceived noise is determined by the maximum sound pressure level (SPL) [8–10]. The heavy-weight impact sound is evaluated according to the sound pressure level (SPL) in the frequency range of 63–500 Hz, and there are various single-number ratings calculated based on the maximum SPL. First, the inverse A-weighted impact SPL ($L_{i,Fmax,AW}$) is calculated by comparing the sound pressure levels in each octave-band from 63 Hz to 500 Hz with those of the inverse A-weighting curve in IEC 61672-1. Furthermore, there are the maximum SPLs to which the fast time constant is applied (L_{Fmax}) and the maximum SPL to which the A-weighted frequency response and fast time constant are applied ($L_{A,Fmax}$) [11,12]. Among them, $L_{A,Fmax}$ most accurately represents the relationship between heavy-weight impact noise and human annoyance

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[13,14]. In addition to the simple SPL, however, the perception of noise caused by floor impact sound has been found to be affected by the spectral, spatial, and temporal characteristics. In particular, the annoyance level is affected by the temporal decay rate, which is determined by the floor and building structures as well as the interior space characteristics [15–17]. Therefore, scientists in many countries have investigated subjective responses to floor impact sound to determine relevant regulation criteria [2,18,19]; hence, grades related to the SPL corresponding to the annoyance percentage (20%, 40%, 60%, and 80%) and based on the dose-response relationship have been proposed. Jeon et al. [2] have assessed satisfaction for various indoor noise sources, including inter-floor noise, through surveys and experiments, and presented grades for each noise source. Further, Jeon et al. [18] have presented floor impact sound grades considering impact factors (such as temporal decay rate and noise sensitivity) in addition to SPL, which were determined in a laboratory environment. However, it remains unclear whether those grades, which are based on test-subject satisfaction and annoyance percentages for noise, can be regarded as the allowance limits of actual residents. In addition, the results of assessments conducted in a laboratory environment are limited as regards understanding of noise-induced annoyance reaction, because noise perception is significantly affected by experience of the actual residence environment.

In this context, virtual reality (VR) technology has recently been utilized to create environments similar to various field situations in the limited laboratory environment [20,21]. When VR technology was used for external environment assessment, the assessors responded more negatively to noises in the virtual space due to the perception of reality provided by the VR environment [22]. Furthermore, the evaluation results were more homogeneous when visual and acoustic stimuli were considered together in a VR environment. Thus, the results of the experiments conducted in the actual environment were very similar to those obtained in the virtual simulation environment [23].

Virtual reality technology can be described as a high-end user interface that includes real-time simulation and interaction through multiple sensorial channels (vision, sound, touch, smell, and taste) [24,25]. The recent studies applying the VR technology to the indoor/outdoor noise evaluation have taken approaches to utilize the creation of three-dimensional sound-field technology and the implementation of virtual space technology [26,27]. An acoustic approach includes sound image localization technology, three-dimensional sound-field creation technology, sound externalization technology, among others [28]. For the sound image localization technology, in which a sound source is placed at the desired position using a headphone, the head-related transfer function (HRTF) can be utilized. Recently, studies have been performed on a sound source calibration method for actual auralization [29] and its application to historical buildings [30]; however, application of this method to assess building noise has not been researched sufficiently. In a study where an environment similar to an actual space was created using a head mounted display (HMD), which is a representative device for virtual space implementation, higher-similarity results were obtained than a case in which an HMD was not used [31]. In addition, the noise-induced annoyance inside a train was assessed to be low using an HMD [32], confirming that the visual information provided by the HMD clearly affected the subjective responses as compared with the results obtained without the HMD. To date, studies have been conducted on the effects of visual and auditory information on subjective responses in terms of noise assessment, but most have focused on external environments [33–36].

In this study, annoyance caused by walking vibration noise, which is a representative noise source associated with complaints for an apartment indoor space, is assessed using VR technology, and regulation grade criteria for heavy-weight floor impact sound are proposed. For this purpose, two types of VR technology are used: (1) HRTF is applied to a single impact sound source to create a directional sound source, and (2) the actual living space for the residents have been modeled and

visual information have been provided using the head-mounted display (HMD). Through psychoacoustic assessment with various SPLs and varying temporal decay rates, the effects of the sound source's directional information and the visual information on the annoyance reaction have been examined. The causes have been investigated by analyzing the physical acoustic parameters. Finally, practical assessment grades for walking vibration noise have been proposed.

2. Methodology

2.1. Heavy-weight impact sound

A rubber ball, which is the standard impact source specified in ISO 10140-3 [4], was used to reproduce the floor vibration and noise caused by continuous walking. The temporal and spectral characteristics of the impact force (1500 N) of a rubber ball are most similar to those caused by running (600–1000 N) or jumping (2000–3000 N) children; this is in contrast to the impact force (4200 N) of the bang machine used in previous studies performed in Japan and South Korea [1,10]. Heavy-weight impact sound was recorded in a reinforced concrete apartment building, which had a box-frame-type structure with various slab thicknesses ranging from 150 to 180 mm. The impact sound was created by dropping a rubber ball in the center of the upstairs living room. The 20-Hz–20-KHz range was recorded without frequency weighting at the center of the living room downstairs at a height of 1.2 m using a 1/2-inch microphone (B&K type 4189) [13]. Among the recorded heavy-weight impact sound sources, the sound source with the frequency characteristics shown in Fig. 1 was used as a representative sound source for psychoacoustic assessment. A sound source with a low background noise level and short reverberation was selected for HRTF synthesis.

2.2. Virtual acoustic modeling

In this study, the noise occurrence scenario was taken as movement of an upstairs resident diagonally across the living room. This is because the movement characteristics of impact sound overhead are the major causes of floor impact sound-induced discomfort for downstairs persons; these sounds occur when children run or adults walk upstairs [15,37]. To assess the annoyance caused by walking vibration noise, the noise level and temporal decay rates (DR) were considered as the

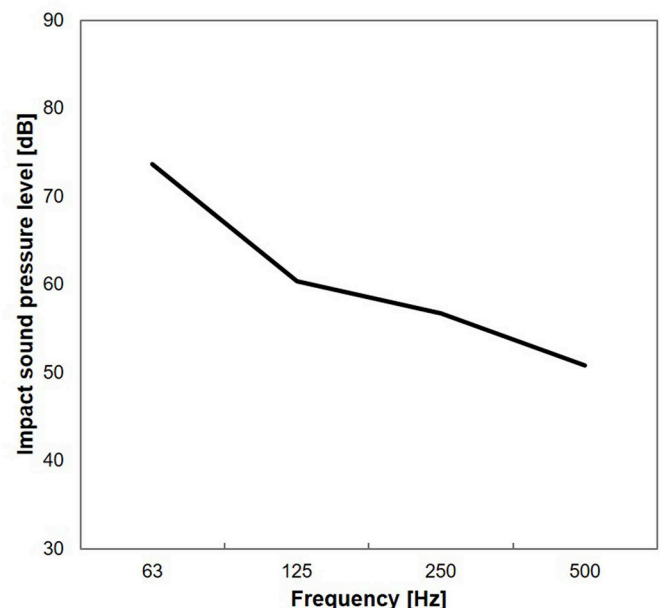


Fig. 1. Frequency response of measured rubber ball sound.

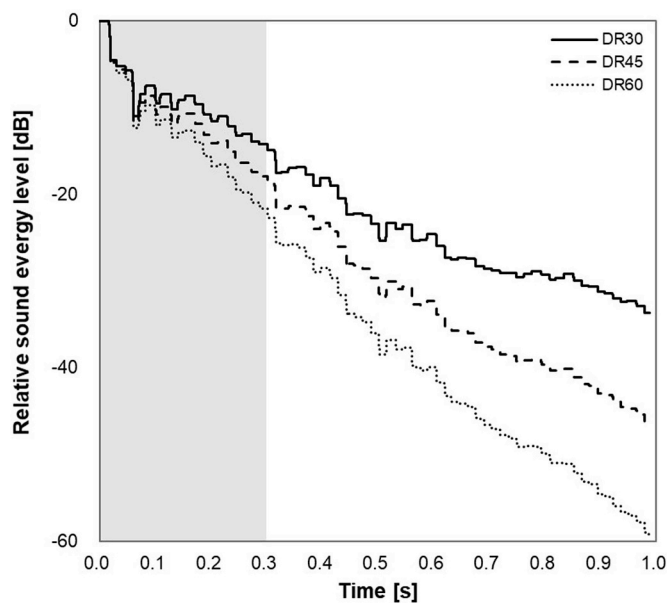


Fig. 2. Temporal decay rates of stimuli.

noise variation factors, and $L_{A,Fmax}$ was used as the single-number rating. The DR was defined as the absolute value of a linear regression line slope in the range of -5 to -35 dB on the normalized Schroeder decay curve of the impact sound, without frequency weighting. Although heavy-weight impact sound is affected by the sound field characteristics (such as reverberation) within the space, the DR was set as a change factor. In addition, various SPLs were targeted, because the DR (dB/s) well represents the temporal decay of heavy-weight impact sound in association with reverberation time.

A total of 27 walking vibration noises were created in a 3 (DR) $\times$ 9 ($L_{A,Fmax}$) configuration. In a previous study, more than 70% of numerous impact sounds generated by an impact ball in apartment buildings were found to be distributed in the range of 30–60 dB/s, and the just noticeable level (JND) was found to be 10.4 dB/s [16]. Therefore, in this study, three DRs were set to sufficiently reflect the typical inter-floor noise occurrence situations. DRs of 30, 45, and 60 dB/s (DR30, DR45, DR60, respectively) were created using the recorded group-A sound source, as shown in Fig. 2. The DR was deformed by equation (1) below:

$$y' = ye^{-nt} \quad (1)$$

where n is a positive number, t is the delay time from the initial point, and y and y' are the amplitudes of the original sound and the modified sound source sample, respectively. For each created DR, nine sound sources were created by increasing the SPL ($L_{A,Fmax}$) from 40 to 64 dB in increments of 3 dBA.

The HRTF is the impulse response due to the sound pathway when a sound travels from the noise source to the eardrum of the test subject, and includes the effects of reflection and diffraction by the human body [38]. In general, humans identify the location of a sound source in 3D space based on the interaural time delay, interaural intensity difference, and interaural spectrum difference. The HRTF is the impulse response including all such information, and directionality can be implemented by convoluting the HRTF to a single-channel recorded source [39]. In this study, the Center for Image Processing and Integrated Computing (CIPIC) HRTF database was used, which contains measurement data for 43 people and two kinds of ear auricles obtained using a KEMAR dummy head [40]. Although measuring and applying the HRTF value for each individual yields more accurate sound image localization performance, in this study, the HRTF data were selected and applied considering the average face size and shape of the test subjects, owing

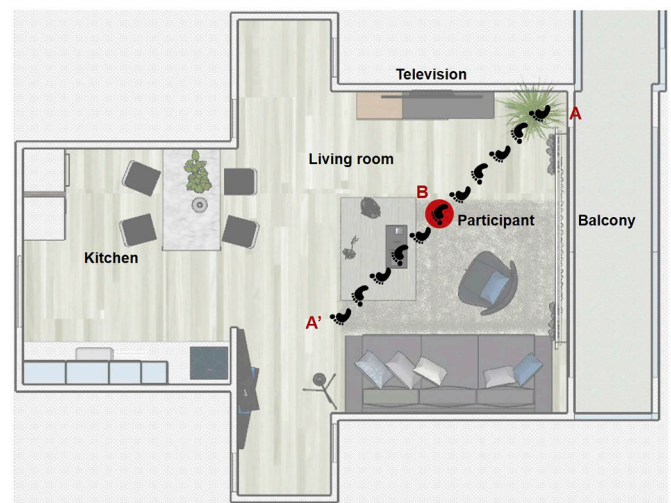


Fig. 3. Impact position for each footstep (A: Start position, A': End position, B: Participant position).

to temporal and technical difficulties. As shown in Fig. 3, while an indoor resident was watching television, it was assumed that the impact sound moved diagonally from the front right-hand corner (A) to the rear left-hand corner (B). To achieve this, the azimuth angle was changed from 45° (A) to 225° , and the HRTF was applied to the single impact sound source so that the elevation could begin at 45° (A) and reach 90° when the impact sound was located above the resident's head (B). The elevation returned to 45° at the last impact position (A').

The time interval between each heavy-weight impact sound source was set to 300 ms and a total of 11 impact sounds were created. To compare the sound pressure levels of the HRTF-applied sound source and the sound source recorded using the mono channel, the sound sources were played on a headphone and recorded using the head and torso simulator (Brüel & Kjær 4100). The sound pressure levels were varied using Audition version 1.5 such that the L_{Amax} of the two sound sources were equal. The frequency characteristics of the HRTF-applied sound and the original sound have been compared and shown in Fig. 4. As an example, the spectrograms before and after HRTF application are presented together for the 50 $L_{A,Fmax}$ sound source of DR30. As apparent from the SPL of each frequency between 63 and 500 Hz, a difference occurred in the SPLs entering both ears, while the major frequency components of the heavy-weight impact sound were maintained. However, slight differences occurred at frequencies exceeding 500 Hz depending on time, but they were less than 1.8 dBA and only minimally affected the psychoacoustic assessment. The SPL change according to distance was analyzed using Odeon software (version 14), and the difference between the maximum and minimum SPL was found to be less than 1.2 dB. As the difference was lower than 1.5 dB, which is the L_{Amax} value obtained for the single impact sound of an impact ball in a previous study [15], no correction for distance was performed.

2.3. Assessment environment setup

To investigate the visual effect on floor impact sound noise assessment through an HMD, a typical apartment living room was modeled using SketchUp 2018 software, as shown in Fig. 5. This model was visualized for use in the HMD using Kubity software [41]. The test-subject eye level was adjusted to 1.2 m to assume a scenario in which the test subject watched television while being seated at the center of the living room.

To investigate the effects of the HMD (visual information) and HRTF (direction information) on the test-subject annoyance and allowance limit for the case of upstairs walking vibration noise in an apartment assessed in the VR environment, four different assessment

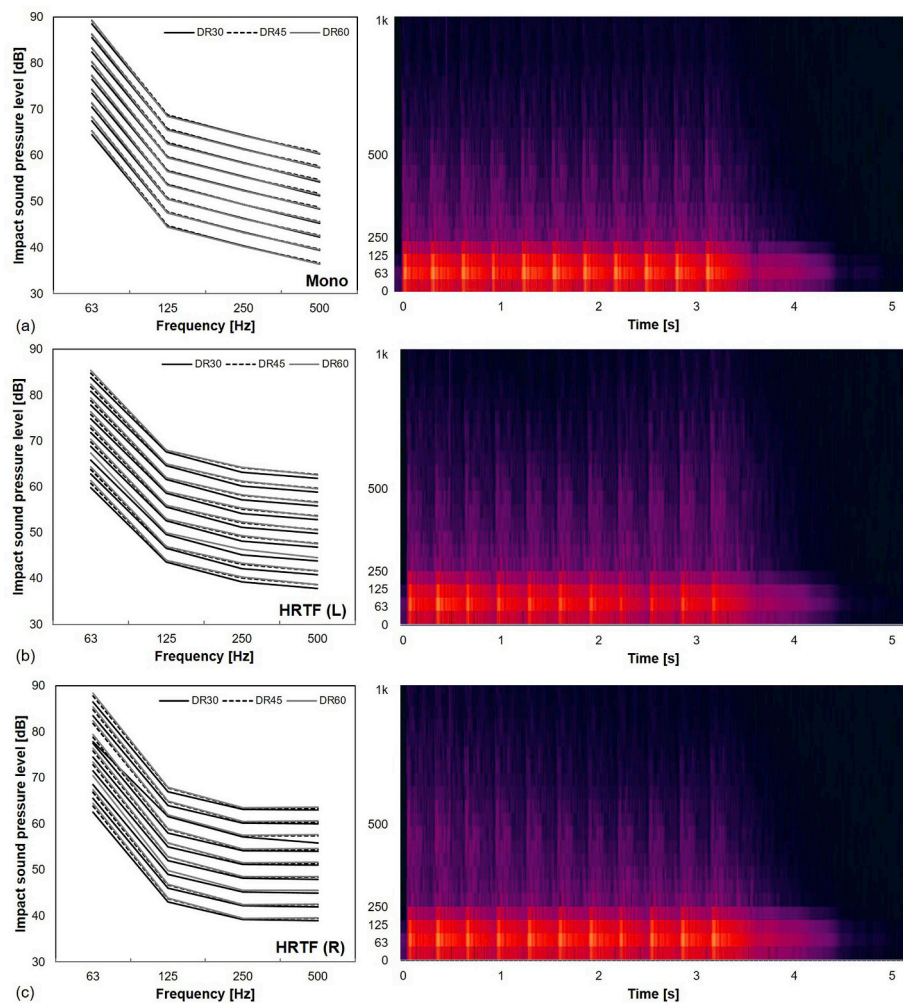


Fig. 4. Frequency responses of stimuli (a) without HRTF, (b) with HRTF (left), (c) with HRTF (right).

environments were set: NONE: a case without the HRTF or HMD; HRTF: a case in which the HRTF only was applied; HMD: a case in which the HMD only was applied; and HRTF + HMD: a case in which both the HRTF and HMD were applied. The experiments were performed in a semi-anechoic room and the background noise was approximately 25 dBA. As heavy-weight impact sound has very clear low-frequency characteristics, use of headphones only is not suitable for psychoacoustic assessment. Therefore, the sound sources in the low-frequency range (< 63 Hz) were reproduced using an open-type headphone (Sennheiser HD-650) and subwoofers (Velodyne DD-10) in the same manner as previous studies [13,16]. In addition, an Oculus Rift 2 (HMD) was used to provide visual information, with high-end computers equipped with G-force GTX 1080 graphic cards or higher being used.

2.4. Procedure

In Korea, the high-density urban policy has developed multi-dwelling housing type such as apartment buildings, and they are mostly wall-type structures. Hence, the floor of the upper unit is connected to the ceiling of the lower unit, and the residents are easily exposed to noise between the floors. A total of 40 Korean test subjects (22 males, 18 females) have participated in the assessment. All of them currently live in a multi-dwelling housing unit and have experienced the walking vibration noise. The test subjects were aged between 20 and 36 (mean age = 24.3, standard deviation = 2.81). Before assessment, the audibility limits of the test subjects were assessed using an audiometer (Rion AA-77). All participants had normal auditory capabilities.

A total of 54 sound sources were used, depending on the SPLs and



Fig. 5. Virtual room modeled with SketchUp 2018.

DRs of the floor impact sound before and after HRTF application. Before the actual psychoacoustic assessment, four types of assessment environments were explained to the test subjects. They were fully informed that they are currently sitting in the middle of the living room watching TV, and a floor impact sound has occurred on the upper floor. Moreover, through a bit of practice, the test subjects were able to understand the assessment method fully. To avoid the test-subject sequential bias, the sound sources were presented in a random sequence. Each sound source had a 4-s duration and was reproduced twice. The interval between the assessment sound sources was 10 s, and the test subjects responded to a survey during this interval. Each test subject performed assessment for all four noise assessment environments. To address the physical inconvenience caused to the test subjects by long-term HMD wear, the HMD and HRTF + HMD assessment environments were not implemented in succession. The maximum experiment time for each assessment environment was less than 15 min, and sufficient rest time was provided.

The test subjects assessed the annoyance level caused by the floor impact sound using the 7-point Likert scale [42–47]. To compare the assessment results with the results of classification of heavy-weight floor impact sounds reported by Jeon et al. [18], the same Likert scale and words for scale points were used for the assessment–7-point Likert scale (0: not at all, 전혀; 1: insignificantly, 별로; 2: somewhat, 조금; 3: moderately, 비교적; 4: considerably, 꽤; 5: highly, 매우; 6: extremely, 엄청나게). Moreover, the word for each scale point was expressed in both English and Korean so the test subjects could clearly distinguish the difference between the scale points. The following question was asked: “If you are in the living room, to what extent are you annoyed by the presented inter-floor noise?” In addition, to investigate the allowance limit for floor impact sound, the following question was asked: “If you are in the living room, can you tolerate the presented inter-floor noise?” The test subjects selected “yes” or “no” as their response. Furthermore, for semantic differential testing of the floor impact sound in order to investigate the test-subject noise perception, 75 words suitable for inter-floor noise assessment were selected from among 600 semantic expressions for the noise experienced in daily life [13]. The 40 test subjects assessed whether the presented word was suitable to describe the floor impact noise using a 5-point numerical rating. Hence, the 12 most commonly selected semantic expressions were determined: “Melodious – Gravelly”; “Calm – Nervous”; “Immersive – Distracting”; “Pleasant – Unpleasant”; “Relaxing – Tense”; “Comforting – Tiring”; “Comfortable – Uncomfortable”; “Static – Dynamic; Orderly – Disorderly; Stable – Unstable; Constant – Variable; and “Steady – Interrupted”. In the experiment, the selected semantic expressions were assessed using a bipolar scale, with the heavy-weight impact sound noise level being fixed at 50 dB ($L_{A,Fmax}$). The selected expressions (+3) were presented on the right-hand side of the questionnaire and their antonyms (−3) were presented on the left-hand side. An assessment of 3 points indicated the strongest expression of feeling for the stimulus.

3. Results

3.1. Annoyance

The annoyance results reported by the test subjects for varying $L_{A,Fmax}$ and DR values for walking vibration noise were investigated, for the different assessment environments. In general, as shown in Fig. 6, the high annoyance level was assessed by the test subjects as $L_{A,Fmax}$ increased, and the coefficient of determination for the linear regression results increased by 0.98 or more on the average. When the Pearson correlation between the changes in both $L_{A,Fmax}$ and DR and the annoyance level was calculated, a significant correlation (R^2 is greater than 0.75, $p < 0.01$) between $L_{A,Fmax}$ and the annoyance level was observed. However, the correlation with the DR proved to be insignificant ($p > 0.05$).

Factorial analysis of variance (ANOVA) was performed to examine

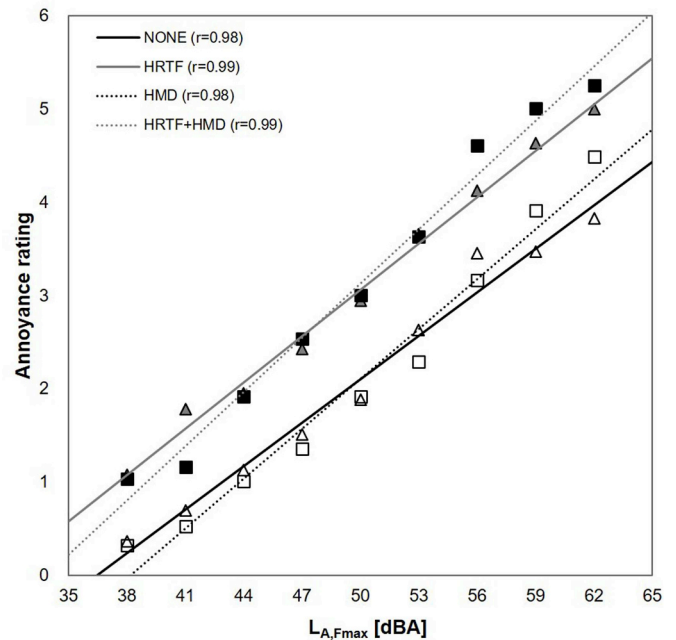


Fig. 6. Mean annoyance ratings as function of SPL (□: NONE, △: HRTF, ■: HMD, ▲: HRTF + HMD).

the significance of the average differences in the test-subject annoyance reactions according to the application of HMD and HRTF, along with the changes in $L_{A,Fmax}$ and DR [16,18,48–50]. This study conducted a Mann–Whitney U test and Kruskal–Wallis test for a nonparametric analysis of the annoyance results, which were evaluated by the Likert scale (See Appendices A and B). The results of the nonparametric analysis showed no significant difference with those of the parametric analysis. Therefore, we described the results based on the parametric analysis. The results are presented in Table 1. An interaction effect of third order or higher indicated significant interactions only. Although the annoyance difference before and after HRTF application was found to be statistically significant ($p < 0.01$), the annoyance difference caused by HMD application was found to be insignificant.

As both the HRTF and HMD had interaction effects with the SPL, the simple main effect of each variable was analyzed to identify the nature of those interactions; the results are presented in Table 2. First, the interaction between the SPL and HRTF shows that there was a significant difference between SPLs regardless of HRTF application, and that annoyance increased with the noise level, as shown in Fig. 6. The cases in which the HRTF were applied were more statistically significant than those in which the HRTF were not applied, regardless of

Table 1

Main and interaction effects of HRTF, HMD, SPL, and DR.

	Independent variable	DF	Mean square	F	p-value
Main Effect	HRTF	1	847.12	926.55	< 0.01
	HMD	1	0.35	0.39	0.54
	SPL	8	906.07	991.02	< 0.01
	DR	2	5.27	5.76	< 0.01
	HRTF × HMD	1	3.06	3.35	0.07
Interaction Effect	HRTF × SPL	8	2.42	2.64	< 0.01
	HRTF × DR	2	1.55	1.70	0.18
	HMD × SPL	8	6.70	7.33	< 0.01
	HMD × DR	2	0.47	0.51	0.60
	DR × SPL	16	1.59	1.74	0.03
	HRTF × HMD × SPL	8	3.328	3.64	< 0.01
	Error	4212	0.91		
	Total	4320			

Table 2
Summary of simple main-effect analysis of data from factorial ANOVA results.

Interaction	Factor	Sum of squares	DF	Mean square	F	p-value
SPL × HRTF	SPL					
	With HRTF	3310.42	8	413.80	466.59	< 0.01
	Without HRTF	3957.45	8	494.68	502.12	< 0.01
	HRTF					
	38 dBA	53.33	1	53.33	109.42	< 0.01
	41 dBA	75.21	1	75.21	119.85	< 0.01
	44 dBA	59.50	1	59.50	85.54	< 0.01
	47 dBA	85.85	1	85.85	101.99	< 0.01
	50 dBA	105.47	1	105.47	110.73	< 0.01
	53 dBA	130.21	1	130.21	108.24	< 0.01
	56 dBA	104.53	1	104.53	82.08	< 0.01
	59 dBA	133.35	1	133.35	110.77	< 0.01
	62 dBA	119.00	1	119.00	104.46	< 0.01
	Error	4026.788	4302	0.936		
SPL × HMD	SPL					
	With HMD	3149.76	8	393.72	346.28	< 0.01
	Without HMD	4152.39	8	519.05	466.44	< 0.01
	HMD					
	38 dBA	1.41	1	1.41	2.36	0.13
	41 dBA	16.13	1	16.13	21.48	< 0.01
	44 dBA	1.30	1	1.30	1.59	0.21
	47 dBA	0.75	1	0.75	0.74	0.39
	50 dBA	0.10	1	0.10	0.09	0.77
	53 dBA	6.08	1	6.08	4.15	< 0.05
	56 dBA	0.83	1	0.83	0.56	0.46
	59 dBA	18.02	1	18.02	12.47	< 0.01
	62 dBA	9.35	1	9.35	86.83	< 0.01
	Error	4839.271	4302	1.125		

SPL. On the other hand, the interaction between the SPL and HMD yielded somewhat complex patterns. Increased annoyance was recorded as the SPL increased regardless of the HMD application. However, the annoyance caused by the use of HMD was higher when the walking vibration noise exceeded 53 dBA. In other words, the test subjects reacted to annoyance more sensitively when visual information was provided through the HMD in an environment with a noise level exceeding a certain level. The main effect of the SPL and DR on the annoyance level was significant ($p < 0.01$). However, when the contribution level of the DR was compared to that of the SPL, it was less than 1% and relatively very low. This result is different from that reported by Jeon et al. (2015), who found that both the DR and SPL affect annoyance, with DR and SPL being 23% and 73%, respectively, when the SPL and contribution were compared. This difference appears to have arisen because the spatial reverberation considered in the present study included only the initial attenuation portion, because the noise source overlapping was limited to the initial 300 ms to reproduce a state of continuous walking. Thus, the test subjects could not perceive practical reverberation differences even though the DR was deformed based on the Schroeder decay curve in this study.

3.2. Allowance limit

Fig. 7 shows the allowance limit results obtained from the test subjects when the level and DR of the floor impact sound were changed in the different assessment environments. The results represent the number of test subjects who replied “yes” to the allowance limit question. As with annoyance, the number of test subjects reporting tolerance decreased as $L_{A,Fmax}$ increased for all assessment

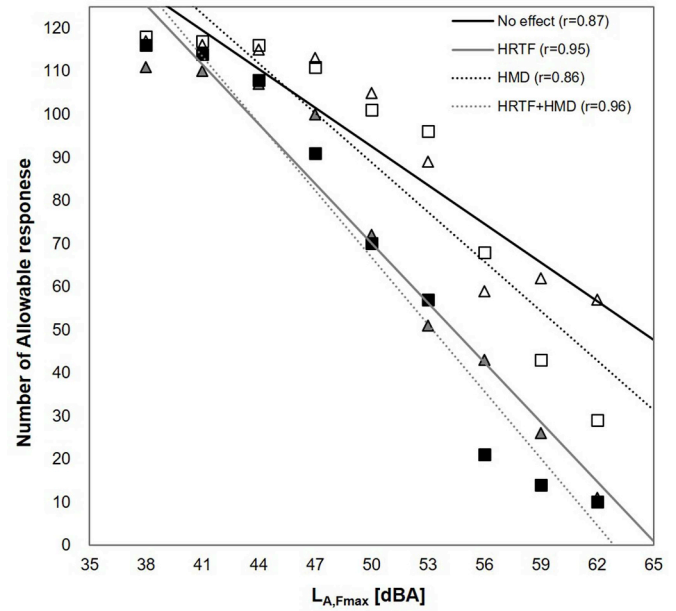


Fig. 7. Number of participant responses indicating allowable noise as function of SPL ($\square$: NONE, $\triangle$: HRTF, $\blacksquare$: HMD, $\blacktriangle$: HRTF + HMD).

environments.

To predict the probability of test-subject noise tolerance depending on application of the HRTF and HMD, logistic regression analysis was conducted. HRTF application, HMD application, SPL, and DR were set as independent variables. The verification results of the entire model compared to the model (constant model) including only the intercept term were statistically significant: $\chi^2(4, n = 4320) = 1705.36, p < 0.01$. The model correctly classified 84.3% of those who responded “yes” to the allowance limit question and 69.2% of those who said “no,” and achieved 79.4% accuracy overall. Table 3 presents the logistic regression coefficient, Wald test, and odds ratio of each independent variable. The HRTF and SPL variables were significant at a statistical significance level of 0.05. The HRTF odds ratio is the ratio of the number of subjects who reported tolerance to the number of subjects who reported lack of tolerance when HRTF was applied, when the other variables were controlled. Here, the odds when HRTF was applied was predicted to be 3.29 times higher than the odds when HRTF was not applied; the HRTF odds ratio was 3.29. In addition, the SPL odds ratio was 1.24, indicating that the odds the test subjects would say “no” to the allowance limit increased by 1.24 times when $L_{A,Fmax}$ increased by 1 dB (when the other variables were controlled).

The univariate analysis results showed that the ratio of those who reported lack of tolerance increased significantly after HRTF application (41.6%) compared to that before HRTF application (24.4%) ($\chi^2(1, n = 4320) = 143.31, p < 0.01$). The group who responded “no” to the allowance limit question (mean = 56.07, standard deviation = 5.67) exhibited a statistically significant low noise level ($L_{A,Fmax}$) compared to the group who responded “yes” (mean = 47.01, standard deviation = 6.83) ($t(4318) = -43.28, p < 0.01$).

Logistic regression analysis for the entire assessment dataset

Table 3

Summary of logistic regression results for allowable noise according to experiment environmental factors and physical acoustic parameters.

Parameter	b	Wald χ^2	p-value	Exp (b)
HRTF	1.19	200.22	< 0.001	3.29
HMD	0.11	1.89	0.169	1.12
SPL	0.21	975.65	< 0.001	1.24
DR	0.00	0.01	0.921	1.00

Table 4

Rotated component matrices of PCA using semantic expression (numbers in parentheses represent explained variance).

Setup	NONE			HRTF			HMD			HRTF + HMD		
Component	1 (40.94)	2 (19.79)	3 (12.22)	1 (45.23)	2 (23.70)	3 (14.01)	1 (40.77)	2 (22.08)	3 (13.69)	1 (46.66)	2 (23.76)	3 (15.35)
Cronbach's Alpha coefficient	0.93	0.81	0.55	0.96	0.92	0.69	0.94	0.86	0.72	0.98	0.92	0.79
Dissatisfaction												
Melodious – Gravelly	0.85	0.18	−0.09	0.89	0.26	0.04	0.78	0.37	−0.13	0.88	0.30	0.06
Calm – Nervous	0.80	0.24	−0.02	0.85	0.20	0.15	0.80	0.25	0.12	0.89	0.24	0.15
Immersive – Distracting	0.81	0.20	0.05	0.87	0.20	0.20	0.86	0.20	−0.19	0.88	0.25	0.07
Pleasant – Unpleasant	0.80	0.23	−0.16	0.87	0.26	0.10	0.86	0.15	−0.10	0.86	0.35	0.10
Relaxing – Tense	0.76	0.29	−0.07	0.79	0.35	0.16	0.80	0.22	−0.04	0.79	0.41	0.24
Comforting – Tiring	0.77	0.32	0.23	0.86	0.34	0.07	0.86	0.21	−0.01	0.84	0.43	0.07
Comfortable – Uncomfortable	0.84	0.37	−0.01	0.80	0.45	0.14	0.74	0.46	−0.06	0.80	0.48	0.15
Irregularity												
Static – Dynamic	0.38	0.64	−0.02	0.33	0.85	0.19	0.28	0.78	−0.08	0.43	0.79	0.21
Orderly – Disorderly	0.33	0.82	0.13	0.38	0.83	0.23	0.27	0.88	0.11	0.37	0.85	0.15
Stable – Unstable	0.24	0.87	0.12	0.32	0.82	0.26	0.28	0.84	0.10	0.47	0.74	0.29
Discontinuity												
Constant – Variable	−0.22	0.18	0.80	0.19	0.30	0.78	−0.01	−0.02	0.89	0.18	0.20	0.87
Steady – Interrupted	0.12	−0.01	0.84	0.07	0.14	0.89	−0.14	0.11	0.86	0.03	0.13	0.91

revealed that the HMD contribution was very low. However, the linear trend line for each assessment environment in Fig. 7 shows that the difference in the HMD-induced annoyance reaction increased as the noise level increased. To statistically examine this phenomenon, logistic regression analysis of the allowance results was conducted while the assessment SPL range was varied. As a result, the HMD effect was found to be significant above 56 dB ($L_{F, Amax}$) ($p < 0.05$), and the verification results of the entire model were also statistically significant ($\chi^2(4, n = 1440) = 140.64, p < 0.01$). Compared to the odds ratio (2.90) obtained for HRTF application, that for HMD application was 1.30, indicating lower influence than the directional information for noise. However, this result also indicates that the visual information on the space has a substantial effect on the noise allowance.

3.3. Semantic expression

Principal component analysis (PCA) using 12 semantic expressions was conducted on the results of the semantic differential test for inter-floor noise; the results are presented in Table 4. The Varimax rotated solution was applied. In the Kaiser-Meyer-Olkin (KMO) test, the KMO measure of sampling adequacy was 0.89 for NONE, 0.91 for HRTF, 0.89 for HMD, and 0.93 for HRTF + HMD, all of which were appropriate levels. In addition, Bartlett's test of sphericity was clear for all assessment environments ($p < 0.01$). As a result of the analysis, the semantic expressions were classified into three factors. Factor 1 corresponded to an explanatory power of more than 40% and had high loading with most words, such as “unpleasant,” “uncomfortable,” “nervous,” and “tiring”; this factor can be interpreted as *Dissatisfaction*. Factor 2 corresponded to an explanatory power of approximately 22%, included “disorderly” and “unstable,” and could be interpreted as *Irregularity*. Factor 3 corresponded to an explanatory power of approximately 14% and had high loading with “variable” and “interrupted” interpreted as *Discontinuity*.

To understand the noise recognition characteristics in the different assessment environments, the factor score for each PCA factor was calculated. Here, the factor score was computed by the regression analysis, which converts variable values belonging to one factor to a factor score [51]. As shown in Fig. 8, the component score distribution differed slightly depending on the assessment environment, confirming that test-subject perception of noise may vary with assessment environment. Compared to the basic environment (NONE), high values were obtained for all component scores for the environments in which HRTF was applied (HRTF and HRTF + HMD). This is because greater dissatisfaction with noise as well as more noise-induced discomfort may be expressed when directional information is provided. Furthermore, component 3 increased after HRTF application, indicating that the test

subjects' perception of sound discontinuity increased.

Table 5 presents the correlations between the semantic expressions in each PCA component and the reported annoyance for the different assessment environments. In the NONE environment, annoyance exhibited positive correlations with factors related to Factor 1 (*Dissatisfaction*), but low correlations with those of Factor 2 (*Irregularity*) and Factor 3 (*Discontinuity*). In the environments in which HRTF was applied (HRTF and HRTF + HMD), the relationship with Factor 1 (*Dissatisfaction*) was clearer than that for the NONE environment. This appears to be because the assessors' perception of “unpleasant,” “tense,” and “uncomfortable” feelings increased significantly compared to the NONE case.

When the HRTF was applied, positive correlations with Factor 2 (*Irregularity*) were observed and a slightly positive correlation with “variable” was also observed. This means that annoyance is assessed differently when the directional information of the sound source is provided; this difference is due to a change in noise discomfort or discontinuity perception, even if the floor impact sound noise level is the same. Compared to the NONE environment, for the environments in which HMD was applied, the correlations between annoyance and the semantic expressions were generally similar; however, a slightly negative correlation with “interrupted” was observed. This indicates that, when the test subjects assessed annoyance in the HMD environment, their responses were relatively muted in relation to changes in sound.

4. Discussion

4.1. Relationship between annoyance and allowance limit

The percentage (%A) of test subjects who reported annoyance caused by heavy-weight impact sound and awarded points of “3: moderately” or higher on the 7-point scale was calculated [18,27,52,53]. This corresponds to a case in which half (50%) the assessors reported dissatisfaction [18]. In addition, the percentage (%AL) of test subjects who responded “yes” to the allowance limit question was calculated. The results are shown in Fig. 9. To correct the subjective assessment data according to the assessment environment, probit regression analysis was conducted. The results are also shown in Fig. 9. Probit regression analysis is a regression model for analyzing binomial response variables, for which a sigmoid curve can be linearized and analyzed using the normal distribution probability [54]. All the probit analysis results exhibited high correlations with the assessment data, with most having correlation coefficients higher than 0.82. The relationship of the floor impact sound SPL, %A, and %AL in the 20–80% section of the probit regression curve was used to represent the linear relationship [2,18,55]. Equations (2)–(5) indicate the linear

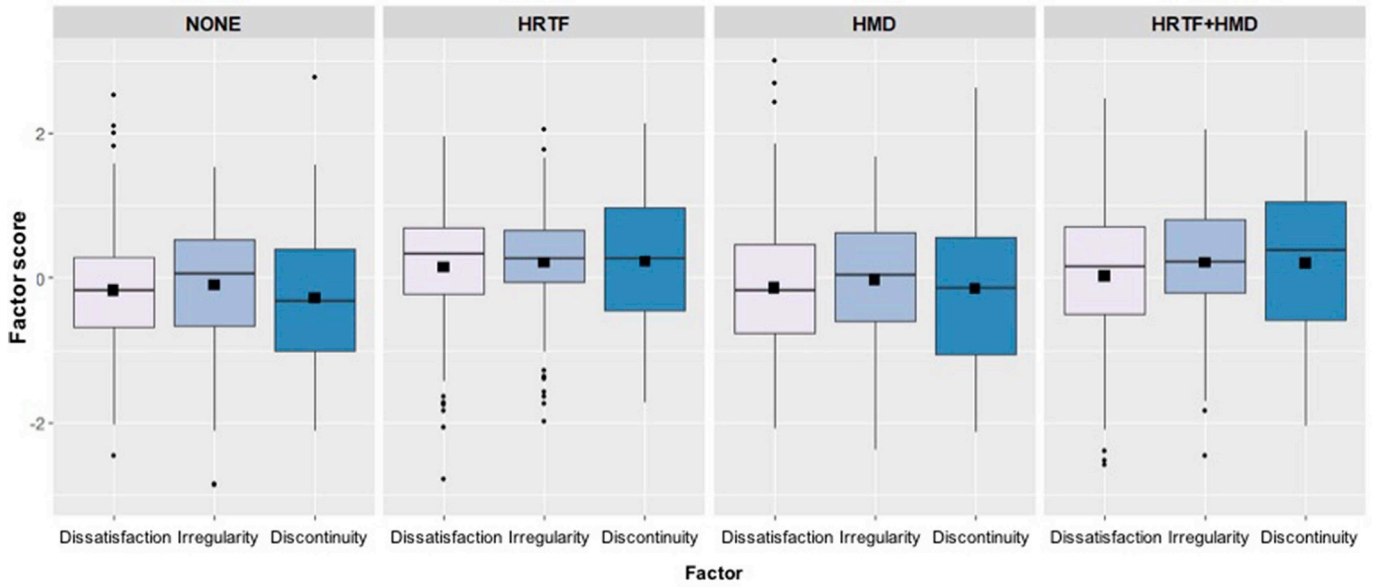


Fig. 8. PCA component scores for semantic factors (dissatisfaction, irregularity, discontinuity) according to experiment setup.

Table 5
Correlation coefficients between semantic expression and annoyance for different experimental setups (**p < 0.01).

Factor	NONE	HRTF	HMD	HRTF + HMD
Dissatisfaction				
Melodious – Gravelly	0.46**	0.57**	0.40**	0.55**
Calm – Nervous	0.45**	0.56**	0.43**	0.53**
Immersive – Distracting	0.49**	0.50**	0.44**	0.52**
Pleasant – Unpleasant	0.30**	0.61**	0.48**	0.56**
Relaxing – Tense	0.29**	0.51**	0.44**	0.55**
Comforting – Tiring	0.43**	0.54**	0.44**	0.58**
Comfortable – Uncomfortable	0.40**	0.62**	0.50**	0.51**
Irregularity				
Static – Dynamic	0.15	0.47**	0.21**	0.42**
Orderly – Disorderly	0.30**	0.43**	0.30**	0.40**
Stable – Unstable	0.12	0.42**	0.26**	0.45**
Discontinuity				
Constant – Variable	−0.15	0.21**	−0.13	0.15
Steady – Interrupted	−0.14	0.12	−0.24**	0.09

relationships with annoyance (%A), and equations (6)–(9) are linear regression equations for the allowance limit (%AL).

$$\%A_{\text{NONE}} = 5.21L_{A,Fmax} - 231.37 \quad (R^2 = 0.99, p < 0.01) \quad (2)$$

$$\%A_{\text{HRTF}} = 5.31L_{A,Fmax} - 200.84 \quad (R^2 = 0.99, p < 0.01) \quad (3)$$

$$\%A_{\text{HMD}} = 5.91L_{A,Fmax} - 265.85 \quad (R^2 = 0.99, p < 0.01) \quad (4)$$

$$\%A_{\text{HRTF+HMD}} = 6.36L_{A,Fmax} - 253.21 \quad (R^2 = 0.99, p < 0.01) \quad (5)$$

$$\%AL_{\text{NONE}} = -3.81L_{A,Fmax} - 276.39 \quad (R^2 = 0.99, p < 0.01) \quad (6)$$

$$\%AL_{\text{HRTF}} = -4.61L_{A,Fmax} - 291.79 \quad (R^2 = 0.99, p < 0.01) \quad (7)$$

$$\%AL_{\text{HMD}} = -5.01L_{A,Fmax} - 336.88 \quad (R^2 = 0.99, p < 0.01) \quad (8)$$

$$\%AL_{\text{HRTF+HMD}} = -5.61L_{A,Fmax} - 339.16 \quad (R^2 = 0.99, p < 0.01) \quad (9)$$

The regression equation slope in the general experiment environment (NONE) was 5.2% per dBA, which is in good agreement with the 5.5% per dBA reported in a previous study [2]. For equations (6)–(9),

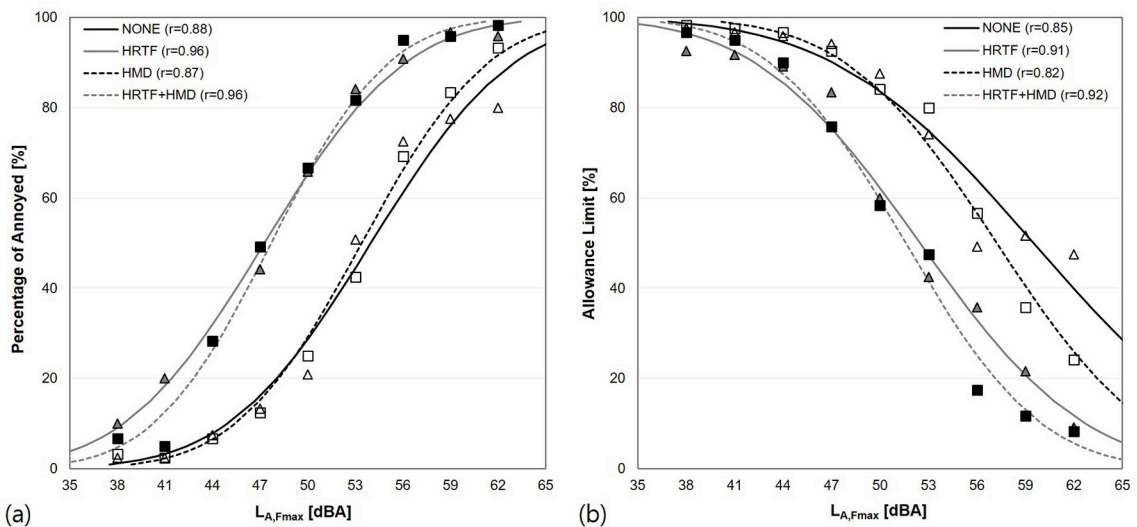


Fig. 9. Percentage of subjects reporting (a) annoyance and (b) allowable noise as function of $L_{A,Fmax}$ for different setups ($\square$: NONE, $\triangle$: HRTF, $\blacksquare$: HMD, $\blacktriangle$: HRTF + HMD).

the slope varied between -3.81 and -5.61 per dBA depending on the assessment environment. For both %A and %AL, the absolute value of the slope increased in the following order: NONE, HRTF, HMD, and HRTF + HMD. This phenomenon was more significant for the allowance limit.

To investigate the relationship between annoyance and allowance limit, the correlation between %A and %AL was analyzed. The correlation coefficient was -0.99 ($p < 0.01$), indicating a highly negative correlation. The allowable noise probability was predicted through logistic regression analysis using the 7-point scale annoyance results as an independent variable. The overall classification accuracy was 86.8%, which was statistically significant: $\chi^2(1, n = 4410) = 2763.60, p < 0.01$. The annoyance odds ratio indicated that, when the annoyance results increased by one point on the 7-point scale, the odds that the test subjects would report a lack of noise tolerance was 4.89 times higher.

4.2. Classification of heavy-weight floor impact noise

Heavy-weight floor impact sounds were graded based on annoyance (%A) for the different assessment environments, as detailed in Table 6. The %A results for the heavy-weight floor impact sounds were divided into 20% intervals and five classes were established. The $L_{A,Fmax}$ value for each section was calculated and the results are presented in Table 6. Class I (0–20%) corresponds to a pleasant noise environment; thus, only less than 20% of the test subjects expressed annoyance in response to floor impact noise. Comparison of the results for the assessment environments indicates that the provision of audiovisual elements through VR technology lowered the satisfaction criterion by 6–7 dB. This result indicates that the test subjects were more sensitive to noise, even if the sound was less intense, when directional information was available. On the other hand, Class V (80–100%) corresponds to a very poor noise environment with high annoyance levels, as more than 80% of the test subjects expressed annoyance or severe annoyance. Note that the floor impact sound grades derived in this study are in good agreement with those derived in a previous study [18], when compared for the same environment (NONE).

The allowance limits at which half (50%) the test subjects reported they could not tolerate the noise were 59.5, 52.5, 57.2, and 51.6 dBA for NONE, HRTF, HMD, and HRTF + HMD, respectively. In other words, as directional and visual noise information was provided, the test subjects exhibited higher noise sensitivity and the ratio of those who reported intolerable noise increased. This finding indicates that it is necessary to implement a residence environment with lower noise than the allowance limit criterion obtained above to satisfy more than 50% of residents. When compared with the grades derived in this study, the allowance limits of each environment were similar to the Class V criteria, indicating that the grade criteria set with %A are significant. In the case of direct impact noise, the inter-floor noise criteria are 57 and 52 dBA for daytime and nighttime, respectively, based on the maximum noise level ($L_{A,Fmax}$ (1 min)) [56]. These values correspond to intermediate grades (Classes III and IV) for the basic experiment environment (NONE), but are associated with the poorest grades (Classes IV

and V) for the experiment environment (HRTF + HMD) in which the most sensitive reactions were observed due to the provision of both directional sound information and visual spatial information. Therefore, differences existed between the existing inter-floor noise criteria and those based on the annoyance felt by test subjects in scenarios similar to an actual apartment environment. This finding shows that improvements to the noise criteria for heavy-weight impact sound are necessary. In this study, HMD visual information was provided for one type of typical living room; however, various floor plans and interiors need to be considered. Moreover, if daytime and night-time environments could be added to the virtual environment by adjusting the level of illumination, a wider range of classification for grade of heavy-weight floor impact noise can be achieved. From an acoustical perspective, further studies, which apply the individual HRTF for each individual or incorporate the indoor sound characteristics based on the physical changes (size, interior finishes, among others) of the interior space, are needed. Furthermore, if head rotation and the change in the position of the test subject in the VR space can be incorporated, a more realistic and accurate grade for heavy-weight floor impact sound can be derived.

4.3. Relationship between acoustic parameters and subjective assessment

To utilize parameters resulting from physical acoustic characteristics to investigate the causes of the subjective assessment results in response to heavy-weight impact noise, auto-correlation function (ACF) parameters were analyzed. That is, ϕ_1 , τ_1 , $VAR(\phi_1)$, and $VAR(\tau_1)$ were analyzed to investigate the noise characteristics related to the perception of pitch sensation; the results are listed in Table 7 [9,37]. All parameter values except for that for $VAR(\phi_1)$ exceeded 0.08 dB (ϕ_1) and 0.4 ms (τ_1), which are the minimum perception levels [57]. Note that ϕ_1 did not differ significantly before and after HRTF application. This appears to be because the test subjects felt that the energies of the main frequency components of the heavy-weight floor impact sound before and after HRTF application were similar, and they perceived no significant change in pitch strength. The τ_1 value decreased after the HRTF was applied and $VAR(\tau_1)$ increased significantly. As τ_1 represents the dominant frequency component (pitch), this decrease in τ_1 means a decrease in the low-frequency components. Therefore, the overall low-frequency sound pressure of the heavy-weight impact sound decreased after the HRTF was applied. In addition, $VAR(\phi_1)$ represents the τ_1 ratio between the initial and tail sections of the floor impact sound [Jeon et al., 2008]. As this parameter exhibited very high values after the HRTF was applied, the low-frequency components were relatively prominent near the tail section despite the decrease in the overall low-frequency sound pressure of the heavy-weight impact sound.

Analysis of the correlations between the annoyance reaction results and ACF parameters revealed that the correlation coefficients with τ_1 ,

Table 6
Classification of heavy-weight floor impact sounds in terms of experimental setup.

Class	%A	$L_{A,Fmax}$ [dBA]				
		NONE	HRTF	HMD	HRTF + HMD	Existing Criteria
I	0–20%	< 48.0	< 41.5	< 48.0	< 42.5	Daytime < 57.0
II	20–40%	< 52.0	< 45.5	< 52.0	< 46.0	
III	40–60%	< 56.0	< 49.0	< 55.0	< 49.0	Night-time < 52.0
IV	60–80%	< 59.5	< 52.5	< 58.0	< 52.0	
V	80–100%	≥ 59.5	≥ 52.5	≥ 58.0	≥ 52.0	

Table 7
ACF parameters for heavy-weight floor impact noise according to HRTF applicability.

Parameter	Statistics	Without HRTF			With HRTF		
		DR30	DR45	DR60	DR30	DR45	DR60
τ_1	Mean	26.54	25.47	24.64	25.06	21.68	20.46
	SD	0.21	0.20	0.67	0.04	0.03	0.39
ϕ_1	Mean	0.29	0.26	0.25	0.29	0.25	0.25
	SD	0.01	0.00	0.01	0.00	0.00	0.00
VAR_{τ_1}	Mean	19.55	34.32	45.2	86.30	140.26	145.16
	SD	4.85	4.37	12.08	0.95	0.90	5.89
VAR_{ϕ_1}	Mean	0.01	0.01	0.01	0.01	0.00	0.01
	SD	0.00	0.00	0.00	0.00	0.00	0.00

$\text{VAR}(\phi_1)$, and $\text{VAR}(\bar{\alpha}_i)$ (except ϕ_1) were -0.32 , -0.30 , and 0.36 , respectively, indicating high statistical significance ($p < 0.01$). Although the effect size is not large when compared to Cohen's conventions on Pearson correlation [58], it can be said that the assessed annoyance can increase somewhat when the overall low-frequency components decrease and the changes in the dominant frequency are small over time.

5. Conclusion

The physical characteristics of heavy-weight impact sound corresponding to annoyance were investigated, and the sound pressure level (SPL, $L_{A,Fmax}$) and temporal decay rate (DR) were varied in a virtual reality (VR) environment to reproduce actual inter-floor noise occurrence scenarios. To implement the scenario in which the floor impact sound source was moving above the subject's head, directional information was provided to both ears over time by combining the head-related transfer function (HRTF) with the mono sound source of the impact sound. In addition, visual information for a typical living room space was provided by modeling a virtual space and utilizing a head mounted display (HMD). Subjective audiovisual assessment was performed for various auditory factors, such as annoyance, allowance limit, and semantic expressions for inter-floor noise. Different experiment environments were employed and the effects of HRTF and HMD, which are key elements in the VR environment, on the subject responses to heavy-weight impact sound were investigated. It was found that the cause of discomfort due to the inter-floor noise occurring in an actual residence space is the change in the directional information for walking vibration noise. Finally, a method for assessing existing grades and a new environment assessment methodology were proposed.

Assessment of the annoyance due to inter-floor noise for different assessment environments revealed that the directional information of the heavy-weight impact sound source affected the results; this information was incorporated via the HRTF. The allowance limit results also confirmed that the HRTF significantly affected noise perception,

because the number of assessors who responded that they could not tolerate certain noises increased by more than three times when the HRTF was applied. It was also found that the visual information of the HMD partially affected the results (by more than 1.2 times) in scenarios with high inter-floor noise (56 dBA). This finding indicates that visual involvement increases as the noise level increases; thus, the influence of the HMD rises. Assessment of the semantic expressions revealed that the downstairs residents expressed causes of annoyance related to floor impact sound in the categories of *Dissatisfaction*, *Irregularity*, and *Discontinuity*. It was confirmed that inter-floor noise occurrence in the HRTF environment causes inconvenience with regard to the building indoor sound environment. This in turn increases the perception of discomfort in the living room environment, thereby increasing resident annoyance caused by walking vibration noise.

The heavy-weight impact sound grades obtained through audio-visual scenario reproduction for inter-floor noise were compared with the results of a previous study [18]; hence, the applicability and necessity of VR technology such as HRTF and HMD for noise assessment in apartment indoor environments was confirmed. This study is significant in that actual environmental characteristics were practically incorporated for auditory assessment to determine grade criteria for inter-floor noise. It is expected that the methodology and results of this study will be effectively utilized to assess the various types of noise encountered in living spaces, because they can reproduce the same effects as actual field situations under limited laboratory assessment conditions.

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Appendix A. Summary of nonparametric analysis with annoyance results according to HRTF, HMD, SPL, and DR

Nonparametric Analysis						
Factor		N	Mean Rank	Nonparametric Analysis		
				Mann–Whitney U	Z	p
HRTF	With	2160	2480.27	1642103.00	−17.122	< 0.01
	Without	2160	1840.73			
HMD	With	2160	2141.54	2291849.50	−1.105	0.31
	Without	2160	2179.46			
Factor		N	Mean Rank	Kruskal–Wallis Test		p
SPL	38 dBA	480	776.56	2604.87		< 0.01
	41 dBA	480	1042.92			
	44 dBA	480	1367.47			
	47 dBA	480	1727.57			
	50 dBA	480	2087.16			
	53 dBA	480	2525.93			
	56 dBA	480	3056.24			
	59 dBA	480	3308.46			
	62 dBA	480	3552.19			
	DR	30 dB/s	1440			
30 dB/s		1440	2197.52			
30 dB/s		1440	2171.28			

Appendix B. Comparison of annoyance with and without HMD according to SPL variations

SPL	Factor	N	Mean Rank	Nonparametric Analysis		
				Mann–Whitney U	Z	p
38 dBA	With HMD	240	228.84	26001.00	−2.012	0.05
	Without HMD	240	252.16			

41 dBA	With HMD	240	216.86	23127.00	−4.007	< 0.01
	Without HMD	240	264.14			
44 dBA	With HMD	240	233.22	27052.00	−1.240	0.22
	Without HMD	240	247.78			
47 dBA	With HMD	240	233.48	27114.50	−1.163	0.25
	Without HMD	240	247.52			
50 dBA	With HMD	240	239.36	28525.50	−0.188	0.85
	Without HMD	240	241.64			
53 dBA	With HMD	240	227.58	25698.00	−2.099	< 0.04
	Without HMD	240	253.43			
56 dBA	With HMD	240	245.86	27514.00	−0.870	0.38
	Without HMD	240	235.14			
59 dBA	With HMD	240	261.40	23783.00	−3.408	< 0.01
	Without HMD	240	219.60			
62 dBA	With HMD	240	254.02	25554.50	−2.211	< 0.03
	Without HMD	240	226.98			

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